Bharathi S

Chennai, India | +91 7010687647 | bharathisenthilkumar28@gmail.com
linkedin.com/in/bharathisenthilkumar28 | github.com/BharathiSen | portfolio

Experience

Software Development Intern Nov 2024
India Meteorological Department – Doppler Weather Radar
- Developed a Python-based terrain-aware radar propagation system for Doppler weather radar site evaluation, processing JAXA DEM/DSM geospatial datasets (GeoTIFF) with GDAL, Rasterio, and GeoPandas
- Implemented beam-blockage analysis using wradlib and visualized radar coverage and terrain effects in QGIS to assess site suitability for radar installation

Machine Learning Intern Aug 2024
Cognifyz Technologies
- Built and shipped end-to-end ML pipelines for prediction, clustering, and recommendation systems, applying feature engineering and hyperparameter tuning with scikit-learn
- Designed data visualization and model interpretability dashboards to communicate AI/ML outputs to non-technical stakeholders; applied anomaly detection on structured datasets

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design
Backend: FastAPI, Next.js API routes, REST design, RBAC/JWT, SSE streaming
Frontend: Next.js 15, React, Tailwind CSS
Data & Infra: PostgreSQL, pgvector, Redis, Qdrant, Prisma, Docker
AI/ML & Retrieval: Agentic AI, RAG pipelines, hybrid retrieval (dense + keyword, RRF), tree-sitter AST chunking, LangGraph, OpenAI/Gemini APIs
Cloud & DevOps: GCP (Cloud Run, Firestore), Cloudflare Workers, Vercel, GitHub Actions CI/CD

Projects

CodeAtlas | *Next.js, FastAPI, Redis, Qdrant, Tree-sitter* Live | GitHub
- Built an AI code-intelligence system that reasons over an entire GitHub repository, not a single pasted file – parses code with tree-sitter into whole-function chunks across 15 languages and answers with cited file/line references
- Engineered hybrid retrieval fusing dense vector search and keyword match via Reciprocal Rank Fusion, and SHA-based incremental indexing that cut re-index time from 7.9s to **0.4s** on unchanged repos
- Implemented a fail-closed Redis quota to bound API spend, automatic fallback to whole-repository context on vector-store failure, and a labelled recall@k/MRR evaluation harness running in CI

Orchestra | *Next.js, FastAPI, PostgreSQL, pgvector, Redis, LangGraph* Live | GitHub
- Built an agentic AI platform where every run is a durable, replayable execution record with per-step cost and latency, spanning three pipelines: direct chat, tool-augmented (planner-tool-reviewer-answer via LangGraph), and multi-agent orchestration
- Implemented an end-to-end RAG pipeline (upload to chunk to embed to pgvector) with a sources UI, and a two-tier memory system (Redis short-term + PostgreSQL long-term) persisting across conversations
- Delivered **0.4–0.9s** direct-chat and **2.3s** warm RAG-retrieval latency through JWT auth, three-axis rate limiting, and SSE streaming

API Reliability Lab | *Next.js, PostgreSQL, Prisma, Redis, OpenAI* Live | GitHub
- Built a full-stack SaaS that load-tests public APIs with 1–20 concurrent requests, streaming live progress over SSE and computing **p50/p95** latency and error rate per run
- Generated AI-powered reliability insights and risk scores via OpenAI with Zod-validated structured outputs, falling back to a heuristic engine when no LLM key is present
- Secured the platform with SSRF-safe validation, SHA-256-hashed API keys, and tiered rate limiting; covered by **22** Vitest unit tests and **5** Playwright e2e tests in CI

Education

Chennai Institute of Technology Chennai, India
B.E. Electronics and Communication Engineering; CGPA: 8.9/10 *2023 – 2027*

Publications

A Literature Review on Cloud–DevOps Synergy for Scalable and Reliable Machine Learning Lifecycle Management, IEEE ICSSS 2025

Proposed the Adaptive Cloud-DevOps Orchestration Framework (ACDOF), reviewing MLOps/Cloud/DevOps architectures to identify scalability, reliability, and sustainability gaps in production ML systems

Certifications

Google Cloud Certified: Cloud Digital Leader | Google Cloud Learning Paths: Cloud Engineering & Generative AI | SQL Programming (SkillRack, Coursera) | CISCO: Programming in C, Networking & Cybersecurity Fundamentals

Problem Solving

- Solved **1,600+** Data Structures & Algorithms problems – LeetCode (331), CodeChef (568), SkillRack (745)